



Startup SG Tech
Project Proposal
Selection Criteria & Guidelines / Template

This document provides a template with a detailed description of how you can approach the task of writing your Startup SG Tech proposal. While it is not strictly necessary to follow the suggested proposal make-up, it would prove useful to do so – the suggested structure attempts to lay out important facets of a project which would portray the project's strengths *in line with the selection criteria*, and therefore help the assessors make a more accurate assessment of the project. Should you feel that other aspects of your project are not sufficiently covered by the suggested structure here, please do include the additional points/aspects in your proposal while keeping to the page limit.

You should note that your proposal will be the only source of information on your project available to the assessors in the early stages of assessment, and therefore, it is of vital importance that you submit a proposal which does justice to your technological idea or concept.

SELECTION CRITERIA

Including, but not necessarily limited to:

1. Strength Of The Scientific / Technical Idea

- Level Of Originality/ Inventiveness/ Novelty
- Technical Feasibility Of Project

2. Ability To Successfully Carry Out The Project

- Scientific / Technical Capabilities Of Development Team
- Business Competencies Of Management Team
- Conceptual Framework, Design And Methodology
- Access To Relevant Resources (Eg. Specialised Laboratories & Equipment)

3. Commercialization Potential

- Potential Manufacturability / Scalability
- Potential To Attract Further Funding & Market Demand
- Defensibility Of Intellectual Property

4. Economic Benefits To Singapore

GENERIC GUIDELINES

Page Limit & Pagination: Your proposal should *not exceed 15 pages, not including* the cover page, references, and supporting enclosures/attachments. Begin numbering your pages after the cover page (at the executive summary page). References and supporting enclosures/attachments may be numbered separately with roman numerals (i, ii, iii, iv, etc).

Page Setting: The proposal should be done on A4 page size setting, with page borders *not less* than 1.2” on the left and 1” on all other sides.

Font: Please use Times New Roman font, *no smaller than* Font size 12.

Startup SG Tech

Project Proposal

BIOMETALLICA

The Applicant shall also make the declaration together with the proposal submission that

- a. The Applicant has complied with all applicable safe distancing and other health measures (“SDMs”) set out in legislation, as well as directions, conditions and advisories issued by the Singapore Government and its agencies, including those issued by MOH (available at www.moh.gov.sg). [Yes/No]
- b. The Applicant agrees to comply with all applicable SDMs that may subsequently be issued by the Singapore Government and its agencies. [Yes/No]
- c. The Applicant agrees that it will allow agencies/entities appointed or authorised by the Agency to access its premises for the purpose of ensuring compliance with the applicable SDMs.
- d. The Applicant understands and agrees that if any of the above declarations are found to be false or inaccurate, or if the Applicant breaches any of the agreements above, the Government has the right to revoke any grants issued, and to immediately recover all monies or other benefits disbursed under the grant.

EXECUTIVE SUMMARY

1. Problem

The global issue of e-waste presents not only an environmental challenge but also a missed opportunity in recycling precious metals, contributing significantly to greenhouse gas (GHG) emissions due to reliance on primary mining. Current recycling methods are inefficient, environmentally damaging, and fail to recover a substantial portion of valuable metals from e-waste and spent auto-catalysts. This inefficiency leads to over \$62 billion in lost value annually, highlighting a critical need for innovative recycling solutions.

2. Solution

BioMetallica proposes an innovative, sustainable approach to precious metal recovery using proprietary bioextraction technology. Our solution leverages genetically engineered microorganisms for bioleaching to efficiently and eco-friendly recycle precious metals from e-waste and spent auto-catalysts. This method drastically reduces environmental impact, operational costs, and energy consumption compared to traditional recycling methods. Our method avoids the use of toxic chemicals, operates at room temperature and produces near zero secondary liquid waste after metal extraction. Our technology is not only environmentally friendly, its 6 times more energy efficient, 3 times cheaper and significantly reduces greenhouse gas emissions compared to traditional precious metal recycling methods.

3. Application

Direct applications of our technology include the recycling of precious metals from e-waste, providing a sustainable source of critical materials for the electronics, automotive, and biomedical industries. Potentially, our technology can be expanded to mine tailings and industrial scrap, broadening the scope of recoverable materials. Successful implementation would revolutionize metal recycling, contributing to a circular economy and reducing dependency on environmentally harmful mining practices.

4. Others

Our proposal directly addresses the urgent need for sustainable practices in material recovery and aligns with Singapore's commitment to environmental sustainability and technological innovation. By advancing recycling technologies, BioMetallica not only proposes a solution to a global environmental problem but also supports Singapore's position as a leader in green technologies and sustainable urban living. Our commitment to R&D, evidenced by our ongoing pilot projects and partnerships, showcases our potential to deliver tangible, impactful results with broad commercial applications.

PROPOSAL NARRATIVE – MAIN TEXT

This document is BioMetallica Pte Ltd's proposal for the SSG POC Grant application. It comprehensively outlines our company profile, team expertise, product innovation, competitive analysis, and a detailed project budget, adhering to the guidelines specified by the SSG POC application format.

1. Company Profile

BioMetallica, a pioneer in sustainable manufacturing, specializes in recycling technologies that transform e-waste and spent auto-catalysts into valuable precious metals. Founded by visionaries and scientific innovators, our mission combats the global e-waste challenge through eco-friendly recovery methods.

Our CEO, Kwadwo "Joe" Konadu Ansah-Antwi, with a Ph.D. in Electrical and Computer Engineering, and Chief Scientific Officer, Li Xirui, a synthetic biology, genetic engineering, and chemistry expert, lead BioMetallica with a dedication to environmental sustainability. This drive is rooted in Joe's firsthand experiences with e-waste in Accra, Ghana.

Having achieved a Technology Readiness Level (TRL) of 6, our validated bioleaching process is set for pilot testing in a 10-liter bioreactor. Our partnerships with BR Metals Pte Ltd and upcoming collaborations for e-waste supply signify our market readiness and highlight our technological commercialization capabilities.

BioMetallica merges innovative metal recovery with sustainability, offering an economically viable, environmentally friendly recycling alternative. This positions us at the forefront of the sustainable recycling sector.

2. Why is Startup SG Tech Support Necessary?

Startup SG Tech's funding is crucial for BioMetallica's aim to transform the recycling landscape in Singapore and globally. Our project on sustainable precious metal recovery from e-waste aligns with Singapore's zero-waste ambitions and environmental goals. Support from Startup SG Tech would accelerate the development of a groundbreaking solution to the pressing issue of electronic waste.

As a technological hub, Singapore faces significant e-waste challenges, which BioMetallica addresses by converting waste into valuable resources. This approach supports Singapore's Green Plan 2030 by minimizing landfill use, reducing greenhouse emissions, and lessening reliance on mining, thus bolstering Singapore's leadership in green technology.

Our technology promises local economic growth through green job creation and positions Singapore as an innovator in sustainable recycling on the world stage. Beyond environmental benefits, BioMetallica enhances public health and offers scalable solutions to waste management across various sectors.

Startup SG Tech backing would mark a strategic investment in Singapore's sustainable development, showcasing a commitment to pioneering a greener, more prosperous future through innovative e-waste management.

3. Detailed POC / POV Project Description

Introduction and Objectives

BioMetallica stands at a critical juncture, advancing a proprietary technology that aims to redefine the recovery of precious metals from electronic waste and spent auto-catalysts. Currently achieving a Technology Readiness Level (TRL) of 6, our project has shown promising results in lab-scale bioleaching, validated within a 2-litre bioreactor environment. The principal aim of procuring the Startup SG Tech grant is to escalate our project from a lab-scale demonstration to a pilot-scale application, illustrating our technology's scalability, economic feasibility, and environmental advantages. The intention is to expand our operational capacity to a 50-litre bioreactor by the project's conclusion.

Scientific / Technological Concept

Our methodology utilizes genetically modified microorganisms for the bioleaching of metals. Targeted integration of the HCN operon has been strategized to surpass natural cyanide production inhibitors, as delineated in Figure 1.

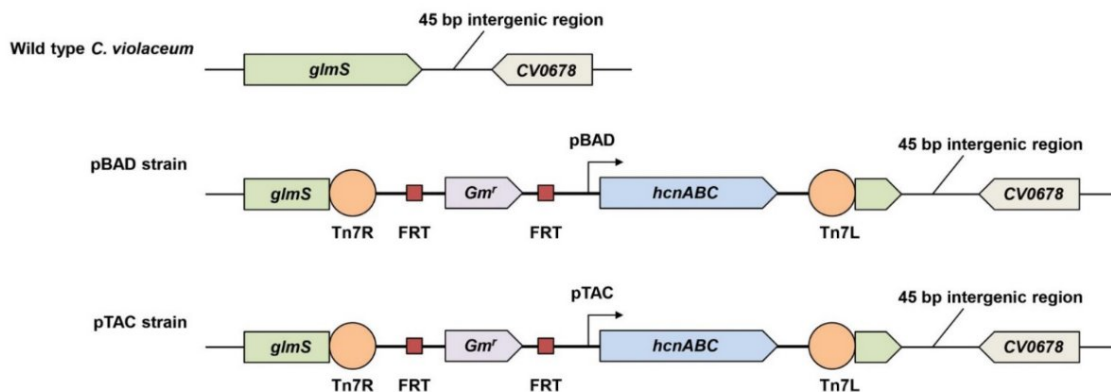


Figure 1 | Lixiviant metabolic engineering of *Chromobacterium violaceum* for enhanced cyanide production and gold recovery. Schematic of genomic region between locus tags CV_0677 and CV_0678 of wild-type and engineered *C. violaceum*. The duplicated *hcnABC* operon, with the respective pBAD or pTAC promoter, was inserted into the *glmS* gene using Tn7-based transposition.

Figure 2 highlights the metabolic pathway of *Chromobacterium Violaceum* (*C. Violaceum*) manipulated to engineer a strain yielding cyanide concentrations of 212 ppm, a significant increase from the wild type's 28 ppm. This bioengineering feat is central to our process designed for the efficient and sustainable recovery of valuable metals such as platinum, palladium, rhodium, silver, and gold.

Contrary to conventional practices that require aggressive chemicals and substantial energy, our method operates at ambient conditions and minimizes environmental impact. The recycling workflow comprises four streamlined stages: pre-treatment, bioleaching, reduction, and separation/precipitation, as illustrated in Figure 3.

Our pioneering pre-treatment technique removes over 90% of non-target metals, enhancing bioleaching yields to 99% for Rh, 95% for Pd, and 92% for Pt.

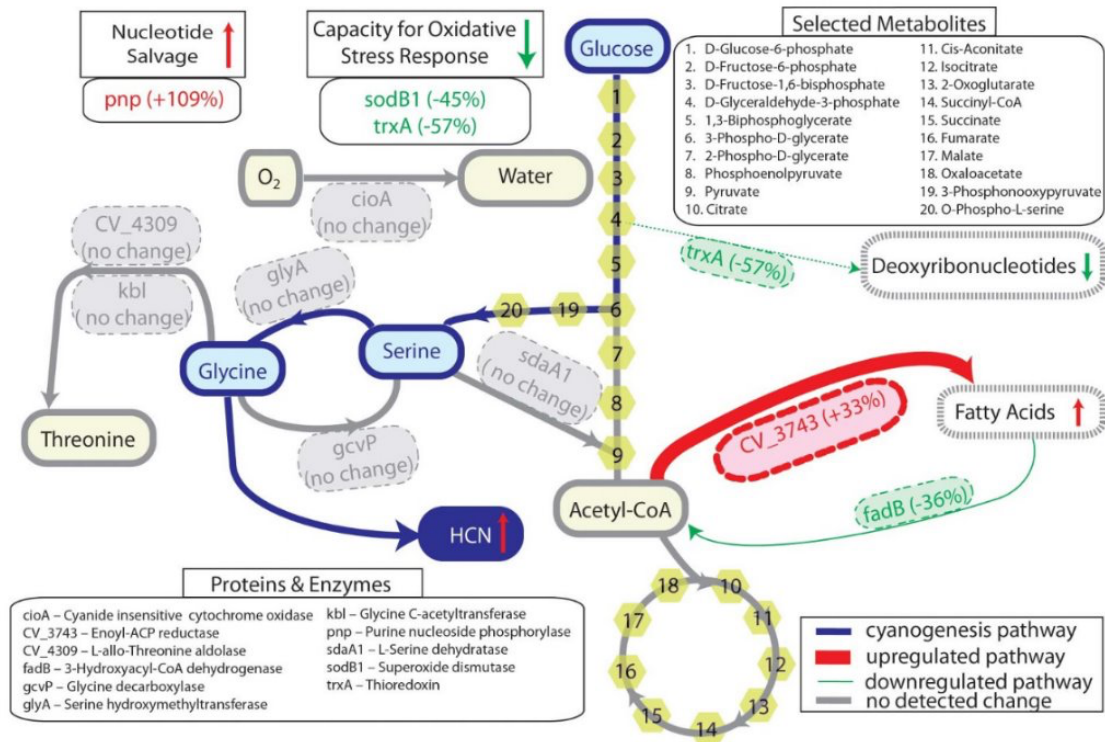


Fig. 2

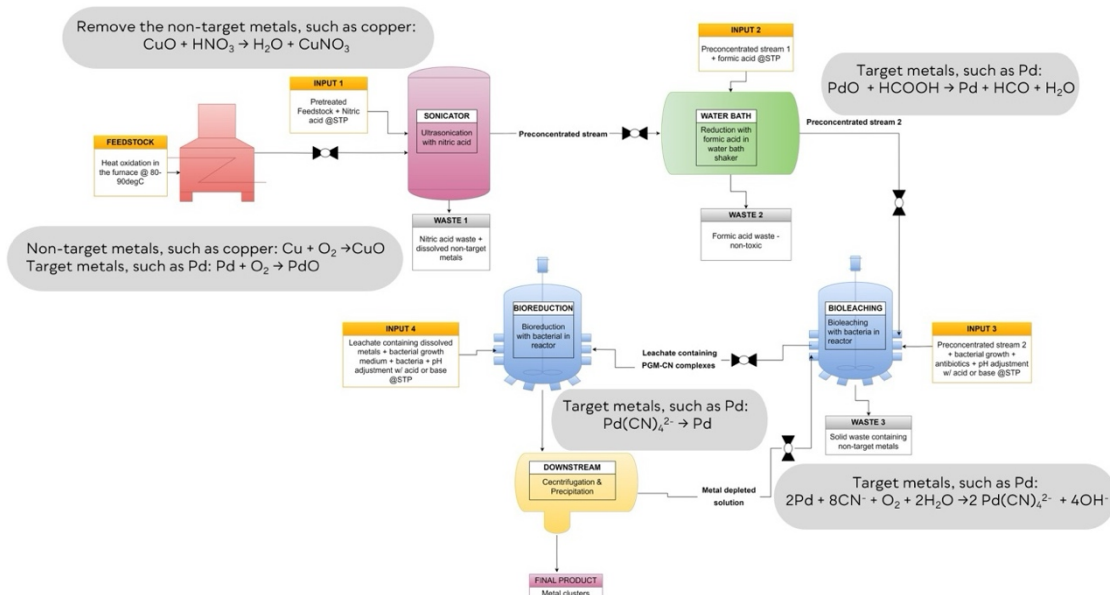


Figure 3: Process Flow Diagram of BioMetallica's bioextraction process

Key Innovations

- **Genetically Engineered Microorganisms:** The employment of *Chromobacterium violaceum* strains, modified to augment cyanide production, boosts bioleaching effectiveness.
- **Optimization of Process Parameters:** Precise management of environmental conditions — temperature, dissolved oxygen levels, particle size, and pH —

- facilitates bioleaching efficiencies exceeding 90% at a solid-liquid mass ratio around 4%.
- **Advanced Pre-treatment Method:** This innovation effectively eliminates over 90% of base metals from the feedstock, mitigating interference in subsequent bioleaching stages.
 - **Intelligent Process Monitoring:** Implementation of IoT (Internet of Things) technologies enables real-time tracking and optimization of the bioleaching operation, ensuring maximal metal recovery as seen in Figure 4.

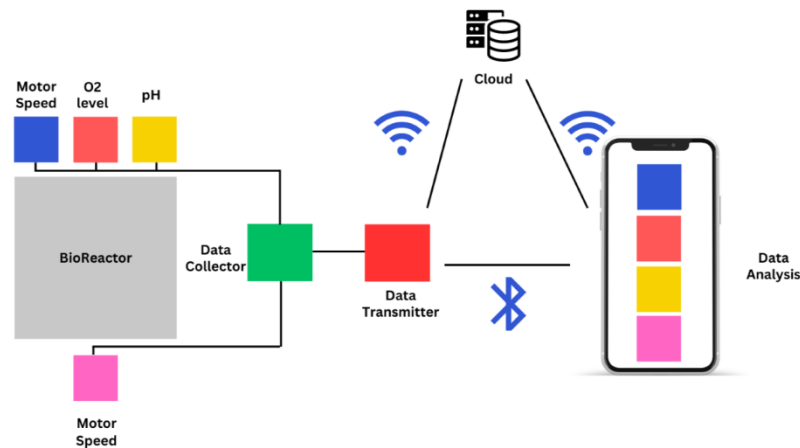


Figure 4

Significance of Innovations

These technological advancements mark a considerable progression from existing metal recovery techniques, introducing a more eco-friendly and economically viable alternative. The innovation in microbial engineering to increase bioleaching efficiency is transformative, potentially diminishing dependence on primary metal extraction methods.

Overall Impact

BioMetallica's technology commercialization could disrupt the precious metal recovery sector, promoting a sustainable alternative to traditional practices. We envision reducing global mining dependency, decreasing greenhouse gas emissions, and resolving the escalating e-waste dilemma. The project envisages setting a new benchmark in the precious metal recovery domain, applicable across the electronic manufacturing and automotive sectors, fostering a transition towards a sustainable, circular economy.

Conclusion

Startup SG Tech's endorsement is pivotal for BioMetallica's progression from conceptual validation to large-scale technological affirmation, highlighting its commercial viability. Our innovations respond directly to the urgent need for sustainable e-waste management solutions, aligning with international environmental objectives and heralding a significant advancement in green technology and recycling methodologies.

4. Novelty of Proposed Approach & Product In Relation to Competitors

BioMetallica's methodology for recovering precious metals from electronic waste and spent auto-catalysts fundamentally diverges from conventional techniques utilized by industry incumbents. Traditional methods such as pyrometallurgy and hydrometallurgy depend on substantial energy inputs and the application of aggressive chemical solvents, respectively. These approaches not only have a high energy footprint but also pose considerable environmental risks through toxic emissions and the generation of hazardous wastes. Entities like Tanaka Holdings and Umicore have largely adopted these conventional practices, which, despite their effectiveness, are increasingly scrutinized for their environmental and energy implications. Emerging entities, including PH7, Mint Innovation, and Phoenix Tailings, present eco-friendlier alternatives but are often restricted by the specificity of feedstock and metal diversity they can process. Specifically, Mint Innovation and Phoenix Tailings leverage microorganisms for selective extraction of gold and rare earth elements, respectively, limiting their feedstock versatility. PH7, targeting platinum group metals similar to BioMetallica, employs solvometallurgy, a process that, while reducing water usage, generates considerable volumes of organic waste.

Bioleaching Innovation:

BioMetallica's bioleaching technology represents a paradigm shift, utilizing genetically engineered microorganisms for the production of biogenic cyanide. This innovative approach allows for the formation of stable metal-cyanide complexes at ambient temperatures, facilitating the extraction of a broader spectrum of precious metals. This method reduces energy requirements and diminishes environmental hazards, positioning BioMetallica at the forefront of sustainable metal recovery technologies. Notably, our process accommodates pulp densities exceeding 4%, surpassing the operational limits of traditional non-hydrometallurgy and pyrometallurgy methods.

Overcoming Historical Limitations:

Previous ventures into bioleaching and alternative eco-friendly methodologies have frequently encountered scalability and efficiency obstacles. BioMetallica's use of engineered microorganisms surmounts these challenges, optimizing bioleaching efficiency and metal recovery at elevated pulp densities. This advancement ensures environmental sustainability and commercial scalability.

Comparative Advantages:

- **Energy Efficiency:** Our processes exhibit a sixfold increase in energy efficiency compared to conventional recovery methods, addressing critical concerns around excessive energy use.
- **Environmental Impact:** Zero wastewater production and reduced chemical reliance establish a new benchmark in minimizing the ecological footprint of metal recovery operations.
- **Scalability and Economic Feasibility:** Demonstrated scalability and a threefold reduction in operational costs position BioMetallica as a viable and cost-effective alternative for industry adoption.

Market Transformation Potential:

BioMetallica's technological breakthroughs offer a compelling, sustainable alternative to existing metal recovery practices. By outperforming traditional methods in key performance metrics such as energy efficiency, environmental preservation, and recovery efficacy, BioMetallica is poised to redefine the precious metal recovery sector. Our technology not

only propels the recycling industry forward but also aligns with global initiatives for a sustainable, circular economy, making a significant contribution to the advancement of green technological solutions.

5. Barriers of Entry & Defensibility

BioMetallica has established multiple formidable barriers to entry, safeguarding its innovative precious metal recovery technology from competitive imitation. These barriers not only solidify our leadership in the market but also ensure the longevity and progressiveness of our operations.

Proprietary Technological Expertise: Central to BioMetallica's competitive edge is our specialized knowledge spanning genetic engineering, microbiology, nanofabrication, metallurgy, and data analytics. The intricate synergy of these scientific fields underpins our proprietary pre-treatment and bioleaching processes, presenting a significant knowledge barrier to replication by competitors lacking in similar interdisciplinary expertise.

Advanced Genetic Engineering: Our strategic application of genetically modified microorganisms for enhanced bioleaching efficiency introduces a substantial technical barrier. These specially tailored strains, exhibiting superior cyanogenic properties, necessitate advanced capabilities in genetic manipulation and a deep understanding of microbial physiology, alongside access to innovative biotechnological facilities for their development and cultivation.

Customized Equipment Design: The bespoke nature of our bioreactors and associated processing equipment, developed through rigorous experimentation and optimization, embodies years of proprietary research. Competitors aiming to replicate our success would need to navigate a complex design and development pathway, requiring considerable time and resources to reach operational efficacy.

Data-Driven Process Optimization: Our competitive moat is further deepened by the extensive dataset we have amassed, detailing operational efficiencies and optimization strategies through sophisticated data analysis and predictive modeling. This repository of knowledge, poised for enhancement through machine learning techniques, sets a high entry threshold for newcomers lacking such foundational data.

Strategic Collaborations and Networks: Strengthening our market position are the collaborative partnerships and networks we have forged with key industry stakeholders and academic institutions. These alliances not only supply us with critical raw materials but also facilitate technological exchanges that are difficult for new market entrants to replicate. Our collaboration with Dr. Donna of the Institute of Materials Research and Engineering (IMRE) in Singapore, aimed at developing an in-situ metal monitoring system using colorimetry, exemplifies our proactive approach to leveraging partnerships for technological advancement.

In summary, BioMetallica's defensibility is rooted in a blend of complex technical expertise, proprietary innovations, and strategic collaborations, creating a comprehensive barrier that ensures our pioneering position in the precious metal recovery sector remains unchallenged.

6. IP Strategy

BioMetallica's intellectual property (IP) strategy is meticulously designed to safeguard our proprietary technologies and reinforce our competitive position in the sustainable precious metal recovery sector. With an acute awareness of the pivotal role IP plays in our enduring success, our approach to IP management is comprehensive, encompassing the rigorous protection, strategic expansion, effective management, and judicious deployment of our IP assets. Currently, we hold exclusive licenses for two critical patents from the National University of Singapore, encapsulating genetic modification protocols for *Chromobacterium Violaceum* and advanced bioleaching processes, including response surface methodologies for process optimization.

Patent Portfolio Development and Protection: At the forefront of our IP strategy is the diligent protection and systematic expansion of our patent portfolio. This effort involves securing our foundational bioleaching patents while actively seeking to patent new discoveries and enhancements. We rigorously evaluate R&D outcomes for patentable innovations, ensuring comprehensive protection across key markets and jurisdictions that are strategically significant for our commercial objectives.

IP Acquisition and Portfolio Management: Collaborating with specialized IP legal experts from the National University of Singapore and City University of Hong Kong, we aim to acquire robust, defensible patents that afford broad coverage for our innovations. Our proactive patent management approach involves regular portfolio assessments to ensure alignment with our strategic business objectives, facilitating informed decisions on patent maintenance, expansion, and potential portfolio optimizations.

Strategic Licensing for Revenue and Partnerships: Our IP strategy incorporates an aggressive licensing model, aiming to monetize our technologies through selective partnerships in ancillary markets. These licensing endeavors are structured to uphold our environmental and quality standards while generating supplementary income, ensuring control over the primary applications of our technology.

IP Defense and Enforcement: Committed to defending our technological advancements and market presence, BioMetallica adopts a robust stance against IP infringement. This includes vigilant market surveillance for potential violations and assertive legal action as necessary. We proactively communicate our IP rights and readiness to enforce them, deterring infringement and underscoring our commitment to protecting our innovations.

Alignment of IP Strategy with Business Objectives: Central to our approach is the seamless integration of our IP strategy with our overarching business goals. This dynamic alignment is continually evaluated and adapted in response to evolving business strategies, technological breakthroughs, and shifts in the competitive and regulatory landscapes.

In essence, BioMetallica's IP strategy is a cornerstone of our business model, crafted to secure and leverage our pioneering technologies in sustainable metal recovery. This strategic framework not only safeguards our innovations but also fuels growth, fosters strategic collaborations, and underpins our long-term leadership in the industry.

7. Commercialisation Plans

Upon successful completion of our Proof of Concept (POC) project, BioMetallica is positioned to transform the precious metal recycling sector with our pioneering, sustainable recovery process. Our commercialization strategy focuses on rapid scale-up, market penetration, and establishing leadership in environmentally friendly metal recovery solutions.

Business Model:

BioMetallica plans to implement a two-pronged commercialization approach, featuring Recycling-as-a-Service (RaaS) for corporate clients and technology licensing for extended reach.

- **RaaS:** Tailored to meet the needs of industries such as precious metal recycling, electronic manufacturing, automotive, and biomedical sectors, this model offers a sustainable, cost-effective alternative to conventional recycling methods. Targeting large firms committed to ESG principles and net-zero objectives, our services promise competitive gross margins of 30-50%, contingent upon the waste stream's metal content and composition.

- **Technology Licensing:** To broaden the impact of our technology, we will license our patented bioextraction processes to global recycling facilities. This strategy not only diversifies revenue but also promotes widespread adoption of eco-friendly recycling practices. A pivotal licensing condition is the allocation of all GHG emission reduction units generated by our technology at licensee facilities to BioMetallica.

Manufacturability and Scalability:

Designed for scalability, our technology features expandable modular bioreactors and established partnerships with engineering and manufacturing firms, ensuring readiness to meet the surging demand for green metal recovery solutions. We aim to establish facilities capable of processing up to 250 tons/year of feedstock, with strategic CapEx investments and calculated payback periods to yield substantial revenue.

Future Funding Prospects:

BioMetallica's groundbreaking technology and the increasing demand for sustainable recycling position us favorably for future investment opportunities. Initial funding from SOSV and forthcoming bridge financing underscore our appeal to investors. Successes from our POC will further attract venture capital, strategic industry investment, and environmental grants, underscoring our scalable impact and investment potential.

Market Validation and Investor Interest:

Our technology's commercial viability is evidenced by collaborations and LOIs from industry leaders like BR Metals Pte Ltd and iZ Precious, Italy. Ongoing discussions with venture capitalists in the green tech sector, including local VC firms like Radical Fund and East VC, reflect strong investor confidence in our innovative recycling solutions.

Supporting Documentation:

Enclosed are the LOIs from BR Metals Pte Ltd and iZ Precious, Italy, alongside summaries of preliminary discussions with venture capital firms, demonstrating the market demand and investment appeal of our technology.

In summary, BioMetallica's strategic commercialization plans are meticulously crafted to ensure the effective market launch and expansion of our sustainable metal recovery technology. By offering an eco-friendly alternative and addressing industry challenges, we are poised to spearhead a more sustainable future in metal recycling.

8. Qualifications of Project Team

BioMetallica's project team is a fusion of technical mastery and business insight, poised to address the intricacies of sustainable precious metal recovery. Our team's diverse expertise spans genetic engineering, chemical engineering, environmental science, and business management, fostering a comprehensive approach to our endeavors.

Leadership and Core Team:

- **Kwadwo "Joe" Konadu Ansah-Antwi, CEO and Founder:** Joe possesses a Ph.D. in Electrical and Computer Engineering, with a focus on renewable energy systems and nanofabrication, augmented by a Graduate Certificate in Management of Technology from the National University of Singapore. His entrepreneurial track record includes the co-founding of a tech startup in the energy storage domain. Joe's synthesis of technical expertise and strategic leadership underpins BioMetallica's direction and innovation.

- **Li Xirui, Chief Scientific Officer and Co-Founder:** With a Ph.D. in Chemistry from Nanyang Technological University, Xirui stands at the forefront of synthetic biology and genetic engineering. Her pioneering work on bioleaching microorganisms is pivotal to our core technology. Xirui's extensive research publications and expertise in metal recovery underscore our scientific and commercial pursuits.

- **Rajeev Kumar, Business Development Advisor:** Rajeev's MBA background and two decades of experience in the chemical and process industries enrich our business development strategy. His prowess in fostering green technology growth and his expansive network is invaluable to BioMetallica's expansion and partnerships.

- **Sam Chow, Financial Advisor:** An ACCA affiliate and entrepreneur in the metal recycling industry, Sam's expertise in finance and sustainability investments supports BioMetallica's fiscal strategy and funding endeavors.

- **Prof Yew Wen Shan, Scientific Advisor:** As the Head of the Chemistry Department at the National University of Singapore, Prof Yew's expertise in genetic engineering and its applications is instrumental. His significant publication record and scientific insights ensure BioMetallica remains at the forefront of the field.

Future Talent Acquisition:

- **Process Engineering Lead:** Targeting a master's or higher in Chemical Engineering with a specialization in bioprocessing or bioleaching, this role is critical for transitioning our processes from lab to commercial scale, emphasizing operational scalability and efficiency.

- **Environmental Compliance Manager:** This position demands a professional versed in environmental science or engineering, with deep knowledge of global sustainability regulations. The manager will ensure our operations align with international environmental standards.

Our team's collective qualifications and experiences, complemented by strategic hires, encapsulate the necessary scientific, technical, and business acumen to propel BioMetallica's mission. Together, we are dedicated to revolutionizing the precious metal recovery sector through innovative and sustainable practices. Further details and the contributions of each team member are available upon request, highlighting our commitment to transparency and excellence.

9. Milestones, Deliverables, Timeline & Decision Points

Task	Milestone	Deliverable	Action By	Due
1	Completion of 50-Litre Bioreactor Design	- Finalize the design of a 50-litre bioreactor for scaled-up bioleaching. - Completed design documents and specifications. - Decision Point: Go/No-Go based on design feasibility and budget estimations.	BioMetallica Pte Ltd	Aug-Sept 2024
2	Pilot-Scale Bioleaching Process Optimization	- Optimize bioleaching process in the 50-litre bioreactor with enhanced microbial strains. - Optimization report detailing process efficiency and recovery rates. - Decision Point: Proceed if recovery rates meet or exceed targeted efficiencies (99% for Rh, 95% for Pd, 92% for Pt).	BioMetallica Pte Ltd	Dec 2024
3	Full-Scale Pilot Testing and Validation	- Conduct full-scale pilot testing of the integrated bioleaching and electrolysis process. - Pilot testing report, including efficiency, environmental impact, and cost analysis. - Decision Point: Evaluate commercial scalability based on pilot results; decide on full-scale commercialization or additional optimization.	BioMetallica Pte Ltd	Mar 2025
4	Commercialization Strategy Development	- Develop a detailed commercialization strategy based on pilot test outcomes. - Commercialization plan, including market entry strategy, partnership models, and financial projections. - Decision Point: Finalize and initiate commercialization strategy if projected ROI and market response are positive.	BioMetallica Pte Ltd	May 2025
5	Launch and Market Entry	Officially launch BioMetallica's technology for commercial use.	BioMetallica Pte Ltd	Jul 2025

		- Commercial launch event, initiation of customer contracts, and operational commencement of service offerings. - Decision Point: N/A - This milestone marks the transition from project completion to ongoing business operations.		
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	Project Months											
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1												
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10. Adequacy of Facilities, Equipment & Resources

BioMetallica's innovative project in sustainable precious metal recovery capitalizes on sophisticated bioleaching and electrolysis technologies, requiring specialized facilities and state-of-the-art equipment. Our strategic resource management aligns with our innovation ethos, guaranteeing the project is well-equipped with the necessary infrastructure and tools.

Existing Resources:

BioMetallica's operations are anchored in a cutting-edge laboratory equipped for chemical processing, featuring a compact bioreactor system and foundational electrolysis setups. The facility is stocked with essential lab apparatus such as autoclaves, centrifuges, and spectrophotometers crucial for microbial culture and metal content analysis.

Equipment Expansion Needs:

To advance our project, we need access to:

- A tailor-made 50-litre bioreactor and related bioprocessing equipment to scale our bioleaching processes.
- A planetary ball mill for fine grinding of feedstock.
- An acid bath setup for initial feedstock processing.
- A BioSafety Level 2 (BSL-2) lab equipped with fume hoods and incubators for secure handling of genetically engineered microorganisms.
- Ultra-low temperature freezers (-4°C and -80°C) for long-term biological sample preservation.
- High-precision analytical instruments for accurate metal quantification, notably an Inductively Coupled Plasma Mass Spectrometry (ICP-MS) system.

Strategy for Resource Acquisition:

- **50-litre Bioreactor:** Design and manufacturing will be outsourced to a specialized engineering firm with bioprocessing equipment expertise.
- **Electrolysis Systems:** Development and assembly of glycol-based electrolysis systems will be conducted in collaboration with our equipment suppliers, ensuring system compatibility and operational efficiency.
- **Analytical Instruments:** Acquisition of ICP-MS and other high-precision analyzers will bolster our metal recovery analysis capabilities.

Collaborative Frameworks:

- **Centers of Innovation and Research Laboratories:** Collaborative agreements with local innovation centers and academic institutions, including the National University of Singapore and the City University of Hong Kong, provide us additional access to advanced equipment and expertise in genetic engineering and electrolysis process optimization.

Conclusion:

Equipped with a blend of internal and collaborative resources, BioMetallica is poised to navigate the complexities of upscaling our technology. The acquisition of advanced

equipment and establishment of key partnerships will amplify our research capabilities, propelling us towards our goal of transforming the precious metal recovery industry with sustainable methodologies.

11. Budget

BioMetallica's project budget is carefully designed to underpin the advancement and scaling of our cutting-edge technology for precious metal recovery. The outlined expenses for the project duration include:

- **Advanced Laboratory Space Lease (BSL-2) in Singapore:** Essential for our expanded operations, including the installation of our 50-litre bioreactor and sophisticated electrolysis systems.
- **New Hires in Singapore (1-2):** Aiming to add a Process Engineering Lead, an Environmental Compliance Manager, and a Laboratory Technician to our team, enhancing our capability in design, development, and testing.
- **Equipment Acquisition and Materials:**
 - Custom-designed 50-litre bioreactor procurement and setup.
 - High-precision metal analyzers for accurate metal quantification.
 - Planetary ball milling machine for optimal feedstock processing.
 - Autoclaves for sterile processing needs.
 - Ultra-low temperature freezers (-4°C and -80°C) for biological sample preservation.
 - Essential laboratory consumables for ongoing research activities.
- **Consultant Fees:**
 - Engaging experts in genetic engineering and bioleaching optimization, and metal separation/purification processes.
 - Legal consultation for the development and management of an IP strategy.
- **Collaboration and Licensing Fees:**
- Costs associated with establishing formal collaborations with academic and research institutions.
 - Licensing fees for integrating external technologies not developed in-house into our process.
- **Marketing and Commercialization Preparation:**

- Investment in market research and strategy development for commercialization, alongside initial marketing initiatives aimed at engaging potential clients and partners.
- **Total Estimated Cost: SGD\$341,300**

This budget underscores BioMetallica's commitment to pioneering sustainable precious metal recovery solutions, ensuring the project is equipped with the necessary resources to achieve its scientific and commercial objectives.

12. Other Supporting Information & Justifications

(Where necessary; ~ 1 page)

An example of scenarios where additional justifications must be provided include cases where foreign professional services have to be chartered, or for R&D to be carried out outside of Singapore.

REFERENCES

(Where applicable; ~ 0.5 page)

SUPPORTING ENCLOSURES / ATTACHMENTS

Please include copies of your latest Financial Statements (balance sheets) and ACRA statements here.

Aside from the abovementioned documents, only supporting information should be included under enclosures and attachments. Eg. Detailed resumes, letters of intent, etc.